Title:
Smart Food Delivery & Order Management System

Based on the case study below, you are required to propose data structure techniques that allows quick retrieval and explain why it is efficient.

Description:
In a Smart Food Delivery & Order Management System, the company GoodTech needs a better way to manage users, restaurants, orders, and delivery services. As a team, you are required to design and implement a system using suitable data structures. First, the system should be able to store and manage customer and restaurant information, allowing data to be added and removed easily. You need to choose appropriate data structures and explain why they are suitable.

Next, the system must process orders in real time, where orders are handled in the order they are received. At the same time, customers should be able to undo the last item added before confirming their order. You are required to design this feature using appropriate data structures and explain your choice.

In addition, the system should assign delivery riders based on priority, such as the shortest delivery time or nearest distance. You need to design a method to select the best rider efficiently and explain how your solution is better than a simple search method.

The system must also find the shortest route between the restaurant and the customer. You are required to represent the locations and implement an algorithm to determine the shortest path, including a brief explanation of how it works.

Furthermore, the system should allow users to quickly search for food items and view them in an organized way. You need to design a structure that supports fast searching and sorting, and explain its advantages.

Finally, the system should provide fast access to important data such as user profiles and order details.

Marking Rubrics:

Component	Data Structure Technique Proposed	Criteria	Marks
User & Restaurant Management	Array / Linked List	Correct DS selection, implementation (add/delete/display), and explanation with complexity	15
Order Processing System	Queue + Stack	Correct use of FIFO (queue) and LIFO (stack), working order flow + undo feature, justification	15
Delivery Assignment	Priority Queue (Min Heap)	Efficient rider selection, correct prioritization logic, performance comparison with linear search	15
Route Optimization	Graph + Dijkstra Algorithm	Correct graph representation, shortest path implementation, explanation of algorithm and complexity	20
Search & Recommendation	BST / AVL Tree	Correct tree usage, insert/search/traversal operations, efficiency explanation (O(log n))	15
Data Retrieval Optimization	Hash Table (HashMap)	Fast data access implementation, key-value usage, explanation of O(1) efficiency	10
System Integration	All DS combined	All modules work together logically and smoothly	5
Presentation & Report	Diagrams (Stack, Queue, Graph, Tree)	Clear explanation, structured report, relevant diagrams	5
TOTAL			**100**